

Comprehensive Clinical Diagnostic Report

Clinical Pathology & Analysis • MediReport AI Diagnostic Engine

11-May-2024
Confidential Medical Findings

PATIENT & CLINICAL INFORMATION

Patient Name:
MISS SYEDA AYESHA D/O ATA U

Age & Gender:
21 Years • Female

Specimen ID:
012336618

Referring Physician:
Not Available

Testing Facility / Lab:
Diagnostic Laboratory, Dhoraji Colony, Karachi

1. OVERALL CLINICAL ASSESSMENT

The patient's urine routine examination reveals pyuria with 10-15 leukocytes/hpf, slightly turbid appearance, trace albumin, and calcium oxalate crystals. These findings are clinically consistent with a lower urinary tract infection (UTI) and mild crystalluria.

2. INFECTION & PATHOGEN SCREENING

Status: Indication of Urinary Tract Infection / Inflammation

Microscopic analysis demonstrates an elevated leukocyte count (10-15 WBCs/hpf) accompanied by slightly turbid urine and trace protein (albumin). This pattern indicates an active immune response to potential bacterial colonization or inflammation in the urinary tract.

3. LABORATORY PARAMETERS & BIOMARKERS

Test Parameter	Result	Reference Range	Status
Leukocytes (WBCs) Elevated white blood cells in urine indicate active urinary tract inflammation or infection (pyuria).	10 - 15 /hpf	0 - 5 /hpf	Elevated
Albumin Trace protein indicates mild irritation or inflammatory exudate within the urinary tract.	Trace	Nil	Elevated
Appearance Turbidity is caused by the presence of excess leukocytes, cellular debris, and calcium oxalate crystals.	Slightly Turbid	Clear	Abnormal
Calcium Oxalate Crystals Presence of crystals in urine can indicate a risk of kidney stone formation if hydration remains inadequate.	(+)	Nil / Absent	Elevated
Epithelial Cells Increased shedding of squamous epithelial cells from the lower urinary tract.	(++) /hpf	Occasional / Few	Elevated
Reaction pH Urine pH is within normal slightly acidic limits.	6.0	4.5 - 8.0	Normal
Specific Gravity Normal urinary concentrating ability of the kidneys.	1.010	1.005 - 1.030	Normal
Glucose No glucose detected, ruling out significant glucosuria.	Nil	Nil	Normal
Red Blood Cells No significant hematuria detected.	Occasional /hpf	0 - 2 /hpf	Normal

4. PHYSIOLOGICAL & PATHOLOGY EXPLANATIONS

Pyuria (elevated white blood cells in urine) occurs when immune cells recruit to the urinary bladder or urethra in response to bacterial infection or mucosal irritation.

Trace proteinuria (albumin) often reflects transient increased permeability or inflammatory damage to the urothelium caused by localized urinary tract infection.

Calcium oxalate crystalluria occurs due to supersaturation of calcium and oxalate ions in the urine, frequently exacerbated by concentrated urine or daily fluid intake.

5. PHYSICIAN RECOMMENDATIONS & ACTION PLAN

Perform a Urine Culture and Sensitivity test to identify the specific pathogen and obtain targeted antibiotic therapy.

Increase oral fluid intake to 2.5–3.0 liters of water daily to help flush out urinary bacteria and dissolve oxalate crystals.

Consult a physician or gynecologist for targeted clinical evaluation and symptom correlation (e.g., dysuria, frequency, urgency).

Disclaimer: This automated diagnostic report is prepared for clinical decision support. Final medical evaluation must be verified by a licensed healthcare professional.